

CHANG DAO, China

WMO: 547510

Lat: 37.9392N

Lon: 120.7256E

Elev: 40

StdP: 100.85

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.9	(c) -5.4	(d) -14.9	(e) 1.0	(f) -4.4	(g) -13.6	(h) 1.2	(i) -3.6	(j) 14.4	(k) -1.5	(l) 13.4	(m) -3.3	(n) 7.2	(o) 340	(p) 0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 4.4	(c) 30.8	(d) 25.3	(e) 29.4	(f) 24.5	(g) 28.3	(h) 23.9	(i) 26.7	(j) 29.6	(k) 25.9	(l) 28.3	(m) 25.2	(n) 27.3	(o) 3.7	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.0	21.4	28.7	25.2	20.4	27.7	24.5	19.5	26.9	84.6	29.7	80.9	28.4	77.5	27.4	30.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.1	(b) 11.4	(c) 10.1	(e) -8.5	(f) 33.3	(g) 2.0	(h) 1.3	(i) -9.9	(j) 34.3	(k) -11.1	(l) 35.1	(m) -12.2	(n) 35.8	(o) -13.7	(p) 36.7		
(4) 13.1	(5) 11.4	(6) 10.1	(7) -8.5	(8) 33.3	(9) 2.0	(10) 1.3	(11) -9.9	(12) 34.3	(13) -11.1	(14) 35.1	(15) -12.2	(16) 35.8	(17) -13.7	(18) 36.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	-0.6	0.8	5.4	11.5	17.2	21.5	24.7	25.1	22.0	16.2	8.9	2.3	(6)
(7)		DBStd	9.59	2.90	3.15	3.64	3.72	3.37	2.77	2.50	1.94	2.16	3.07	3.83	3.13	(7)
(8)		HDD10.0	1069	329	258	151	24	0	0	0	0	2	66	241		(8)
(9)		HDD18.3	2616	587	491	401	206	63	6	0	0	2	77	284	498	(9)
(10)		CDD10.0	2159	0	0	8	70	224	346	457	468	359	195	32	0	(10)
(11)		CDD18.3	663	0	0	0	2	28	101	199	209	112	12	0	0	(11)
(12)		CDH23.3	3711	0	0	0	7	112	465	1404	1405	309	7	0	0	(12)
(13)		CDH26.7	731	0	0	0	1	14	84	357	256	21	0	0	0	(13)
(14)	Wind	WSAvg	4.9	5.7	5.2	5.2	5.3	4.8	4.1	3.9	3.8	4.1	5.0	5.9	6.0	(14)
(15)	Precipitation	PrecAvg	560	6	10	12	25	49	60	165	124	50	25	23	11	(15)
(16)		PrecMax	882	19	36	76	76	119	149	554	248	134	74	81	34	(16)
(17)		PrecMin	259	0	0	0	0	6	5	20	10	8	0	0	0	(17)
(18)		PrecStd	150	6	10	15	17	28	38	121	64	34	19	19	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	11.5	18.5	24.1	28.4	30.7	33.1	32.3	28.7	24.4	19.2	11.4	(19)
(20)			MCWB	3.8	6.4	11.0	15.4	18.5	21.5	26.3	27.0	23.7	18.6	13.7	8.7	(20)
(21)		2%	DB	5.6	8.6	15.1	21.5	26.0	28.9	31.4	30.3	27.1	22.7	16.8	9.2	(21)
(22)			MCWB	3.1	4.9	9.1	14.0	17.3	20.6	26.0	26.1	22.5	17.9	12.4	6.6	(22)
(23)		5%	DB	4.3	6.5	12.6	19.3	24.3	27.4	29.9	29.0	26.1	21.4	15.5	7.6	(23)
(24)			MCWB	2.0	3.4	7.5	12.5	16.6	20.4	25.4	25.3	21.8	17.0	11.9	5.1	(24)
(25)		10%	DB	3.2	5.1	10.6	17.4	22.6	26.0	28.6	28.0	25.1	20.2	14.1	6.4	(25)
(26)			MCWB	1.1	2.4	6.1	11.5	16.0	20.0	24.6	24.8	21.0	16.3	10.6	3.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.1	7.1	11.6	16.9	20.2	24.2	28.2	27.8	25.2	20.3	14.9	9.1	(27)
(28)			MCDWB	6.3	10.1	17.0	21.9	25.9	27.9	31.0	30.7	26.9	22.7	17.6	10.5	(28)
(29)		2%	WB	3.6	5.2	9.7	14.9	18.8	22.6	26.9	26.9	23.7	19.1	13.6	7.0	(29)
(30)			MCDWB	5.1	7.9	14.3	19.6	24.3	26.5	30.1	29.5	26.2	21.2	15.9	8.6	(30)
(31)		5%	WB	2.5	3.9	8.1	13.5	17.7	21.7	26.0	26.1	22.7	18.1	12.3	5.5	(31)
(32)			MCDWB	4.1	5.9	11.9	18.3	22.5	25.5	28.8	28.3	25.1	20.6	15.1	7.3	(32)
(33)		10%	WB	1.3	2.8	6.6	12.1	16.7	20.9	25.2	25.4	21.9	17.0	11.1	4.3	(33)
(34)			MCDWB	2.9	4.6	10.0	16.6	21.4	24.4	27.8	27.3	24.2	19.7	13.7	6.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.8	4.6	5.6	6.7	6.6	6.3	5.2	4.4	4.5	4.6	4.3	3.8	(35)
(36)			MCDBR	5.3	6.8	8.5	9.8	8.8	8.3	6.9	5.5	5.5	5.7	5.9	5.1	(36)
(37)			MCWBR	4.1	4.8	5.3	5.5	4.8	3.8	3.3	3.0	3.3	3.8	4.2	4.4	(37)
(38)		5% WB	MCDWB	4.9	6.4	8.2	9.3	7.9	7.0	6.1	5.2	4.9	5.3	5.6	5.0	(38)
(39)			MCWBR	4.3	4.9	5.4	5.7	4.8	3.8	3.4	3.0	3.4	4.1	4.8	4.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.477	0.532	0.575	0.617	0.647	0.742	0.729	0.658	0.582	0.601	0.550	0.492	(40)	
(41)		taud	1.851	1.760	1.703	1.673	1.658	1.543	1.617	1.739	1.830	1.704	1.767	1.841	(41)	
(42)		Ebn at Noon	644	665	687	692	682	619	622	655	677	595	565	590	(42)	
(43)		Edh at Noon	153	190	221	240	248	278	257	222	192	198	166	145	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.23	3.06	4.39	5.20	5.76	5.43	4.80	4.68	4.29	3.30	2.32	1.90	(44)	
(45)		RadStd	0.20	0.28	0.33	0.24	0.45	0.44	0.48	0.47	0.33	0.28	0.29	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	(47) Regional (5 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
										+82	+48

Nomenclature: See separate page